



Topics in Cognitive Science 8 (2016) 222–241
 Copyright © 2015 Cognitive Science Society, Inc. All rights reserved.
 ISSN:1756-8757 print/1756-8765 online
 DOI: 10.1111/tops.12177

Climate Projections and Uncertainty Communication

Susan L. Joslyn, Jared E. LeClerc

Department of Psychology, University of Washington

Received 30 June 2014; received in revised form 2 April 2015; accepted 31 October 2015

Abstract

Lingering skepticism about climate change might be due in part to the way climate projections are perceived by members of the public. Variability between scientists' estimates might give the impression that scientists disagree about the *fact* of climate change rather than about details concerning the extent or timing. Providing uncertainty estimates might clarify that the variability is due in part to quantifiable uncertainty inherent in the prediction process, thereby increasing people's trust in climate projections. This hypothesis was tested in two experiments. Results suggest that including uncertainty estimates along with climate projections leads to an increase in participants' trust in the information. Analyses explored the roles of time, place, demographic differences (e.g., age, gender, education level, political party affiliation), and initial belief in climate change. Implications are discussed in terms of the potential benefit of adding uncertainty estimates to public climate projections.

Keywords: Uncertainty; Risk perception; Climate change; Decision making; Judgment; Experimental psychology

1. Climate projections and uncertainty

Despite the fact that there is nearly complete consensus among climate scientists that the global climate is changing and that human activity is the cause (Cook et al., 2013), a substantial minority of Americans remain skeptical. A recent survey in the United States found that 36% of those polled either did not believe in climate change or were unsure (Leiserowitz, Maibach, Roser-Renouf, Feinberg, & Howe, 2013). Lingering skepticism

Correspondence should be sent to Susan L. Joslyn, Ph.D., University of Washington, Department of Psychology, Box 351525 Seattle, WA 98195. E-mail: susanj@uw.edu

may be at least partly to blame for the fact that large-scale action to mitigate climate change has as yet failed to occur.

The reasons for the remaining skepticism are undoubtedly numerous and complex (Center for Research on Environmental Decisions [CRED], 2009). One potential contributing factor is the impression that climate scientists themselves are conflicted (Ding, Maibach, Zhao, Roser-Renouf, & Leiserowitz, 2011). A recent survey revealed that 33% of American respondents believe there is “a lot” of disagreement about global warming among scientists (Leiserowitz et al., 2013). This misperception may stem from the considerable uncertainty regarding the precise extent, timescale, and consequences of climate change. In some cases, the best scientific data support a wide range of values. Skeptics might misinterpret this as lack of scientific consensus about the *fact* of climate change (Freudenburg & Muselli, 2010; Smithson, 1999). If so, including a numeric likelihood estimate might serve to clarify that much of the variability is due to quantifiable uncertainty inherent in the prediction process rather than to fundamental differences in scientific opinion about global warming. Uncertainty estimates might increase the public’s trust in climate change projections.

Indeed, there is a growing literature suggesting that uncertainty information in weather forecasts enhances trust in the forecasts and improves decisions (Nadav-Greenberg & Joslyn, 2009; Roulston, Bolton, Kleit, & Sears-Collins, 2006). For instance, in one group of experiments (Joslyn & LeClerc, 2012), participants made a series of decisions about whether to spend money from a set budget to apply salt brine to the roads during the winter to prevent icing. If they failed to salt and freezing temperatures were observed, a substantial penalty was deducted from the same budget. The goal of the task was to maximize the budget: Participants should apply salt only when they thought it necessary. Participants based their decisions on an overnight low-temperature forecast. Some participants saw only nighttime low temperatures, whereas others also saw the percent chance of freezing. Participants with the chance of freezing had greater trust in the forecast and made significantly better decisions than did participants with only the nighttime low temperature. This may have been because forecasts that included uncertainty information seemed less wrong when the observed temperature did not exactly match the single-value forecast. Probabilities may also have added information critical to improving decisions. Moreover, probabilistic forecasts may have better aligned with people’s intuitions gathered over extensive pre-experimental experience with forecasts and outcomes (Joslyn & Savelli, 2010; Morss, Demuth, & Lazo, 2008; Savelli & Joslyn, 2012). Therefore, forecasts that acknowledge the uncertainty explicitly might seem more realistic, trustworthy, and complete.

Including uncertainty information in climate projections may have a similar effect. Indeed, the International Panel on Climate Change, a key source of climate change information, provides explicit advice to its contributors regarding the use of uncertainty language. However, little is known about the impact of uncertainty on trust in climate projections. Uncertainty information may make climate projections seem more trustworthy. This may in turn inspire greater concern that translates into mitigative action. However, there are some key differences between climate projections and weather forecasts.

Most notably, climate projections concern far-future events, in many cases beyond the lifetime of the user. Thus, if the benefit of uncertainty in weather forecasts depends on comparing the forecast and outcome, it might not extend to climate projections. In addition, there is concern among scientists themselves that acknowledging uncertainty will have the opposite effect, making scientists appear incompetent (Frewer et al., 2003; Shackley, Risbey, Stone, & Wynne, 1999), or enhancing the perception of disagreement among scientists. Moreover, climate change is a complex and controversial issue and attitudes toward it may be affected by demographic variables such as gender, age, education level, and political affiliation (e.g., Budescu, Por, & Broomell, 2012; Hart & Nisbet, 2012; Lewandowsky, Gignac, & Oberauer, 2013; McCright & Dunlap, 2011). As such, uncertainty information may serve to bolster existing skepticism among some groups, decreasing rather than increasing trust (Hart & Nisbet, 2012). In short, including uncertainty information in climate projections could have either a positive or negative effect on trust that may depend on demographic characteristics.

One of the major goals of the research presented here was to test these questions. The two experiments described below compared the impact on trust of including uncertainty information in climate projections. Experiment 2 also tested how uncertainty information interacts with demographic characteristics.

The other major goal of this research was to explore the impact of climate communication on concern. Even those who trust climate projections may be insufficiently concerned to motivate climate-friendly behavior for other reasons. In a recent poll in the United States (Leiserowitz et al., 2013), less than half of those surveyed reported that climate change poses an imminent threat. Only 14% of respondents indicated being “very worried” about climate change. This may be due at least in part to how climate projections are expressed. For instance, most climate projections describe global averages, abstractions that might seem irrelevant to most people and fail to generate an affective response (Weber, 2006). If so, local projections (e.g., sea level rise in one’s own region) might generate more concern (Leiserowitz, 2006) because they may be more personally relevant.

However, some factors that may detract from concern are inherent in climate change. For instance, many of the most serious consequences will occur in the distant future and may be less worrisome (Ainslie, 1975; Hardisty & Weber, 2009) due to a psychological process known as temporal discounting, assigning less value to delayed outcomes as compared to immediate outcomes (Chapman, 2002). Some research (e.g., Hardisty & Weber, 2009) suggests that under certain circumstances environmental and monetary discount rates are similar, supporting this hypothesis. However, other research (Baron, 2000) suggests less discounting for environmental goods than for money (Baron, 2000). Low discounting for environmental outcomes might occur because decision makers recognize that the well-being of the environment is equally important in the present and future or because environmental changes are typically evaluated as losses, which can lead to lower discount rates than gains (Hardisty & Weber, 2009). In addition, concern, like trust, might be influenced by demographic characteristics such as gender, age, education, or political affiliation. Therefore, Experiment 1 described below tested the impact of time frame and

whether the outcome was global or local on concern. Experiment 2 tested the relationship between demographic characteristics and concern.

Both experiments used a questionnaire that presented climate projections for temperature and sea level and asked participants to give trust and concern ratings for each. Participants also provided their own numeric estimates, which constitutes an indirect assessment of respondents' trust in the climate projections. For example, given a projection of a 3°F (1.67°C) increase in average global temperature, some people might think there will be no increase in global temperature (0°F), suggesting that they do not believe in global warming at all. Others might think the increase will only be 2°F (1.11°C), suggesting that they regard the projection as an exaggeration.

In both experiments, the communication format was manipulated between participants. Some participants were shown only single-value projections (e.g., temperature increase of 3°F [1.67°C]), to which we will refer as deterministic projections, implying a single outcome; other participants were also shown predictive intervals within which the observed value was expected with a 90% probability.¹ A pilot experiment, which served to refine our procedures, was conducted on university student participants. The two experiments reported below were administered on the World Wide Web in populations with a wide range of demographic characteristics.

2. Experiment 1: Weather blog participants

In Experiment 1, three variables were manipulated between groups: time frame (target year), whether the outcome was global or local (place), and whether a probabilistic or single-value estimate was included (projection format). The temperature and sea level projection values were the same across target years and place. That is, all participants were shown the same values, but the stated target year was systematically manipulated and for half of participants the outcome was described as local and for the other half as global. This allowed for direct comparisons between the levels of the independent variables of interest (target year, place) without introducing a confound (different projection values).

We predicted greater trust and concern for near- than far-future projections due to temporal discounting and greater trust and concern for local than for global projections due to personal relevance. Trust and concern may also be affected by format. If uncertainty signals disagreement or incompetence among climate scientists, then participants may express less trust and concern when receiving a probabilistic projection. If, however, participants have intuitions about the uncertainty inherent in climate projections, then probabilistic projections may seem more trustworthy and lead to greater concern.

2.1. Method

2.1.1. Procedure

The questionnaire was posted on a popular Pacific Northwest United States weather blog for approximately 2 weeks in August 2012. Participants provided information

about their age and gender and then completed the questionnaire. They were not compensated.

2.1.2. *Participants*

A total of 818 people, 30.2% female, ranging in age from 18 to 84 (mean = 49.16 years), completed the questionnaire described below. Ninety-three percent of participants were from the Puget Sound region in Washington State.

2.1.3. *Questionnaire*

The questionnaire included two climate projections described as increases over the 20th century average: average temperature increase of 3°F (1.67°C) and average sea level rise of 6 inches (15.24 cm). Half of questionnaires also included 90% predictive intervals (temperature: 2–4°F [1.11–2.22°C], sea level rise: 3–9 inches [7.62–22.86 cm]). These same values were used for all target years (2015, 2025, 2050, 2100) and for both global and local projections. Note that temperature and sea level values shown to participants were in degrees Fahrenheit and inches, respectively; the equivalents in degrees Celsius and centimeters reported here are for readers' convenience. The projections were actually global values for 2050 and based on climate reports (e.g., IPCC, 2007), the NOAA website, and the MAGICC/SCENGEN program, a computer program that customizes projections based on different climate models.). The pilot study used actual varying projections (increasing by year and greater for Global averages), but the results were similar. After reading each projection, respondents answered five questions pertaining to it. The first asked them to provide their own single-value estimate (e.g., "I think the average global temperature in 2050 will be ___°F"), the second asked for an upper bound estimate (e.g., "I would not be surprised if the average global temperature in 2050 were as high as ___°F"), and the third asked for a lower bound estimate (e.g., "I would not be surprised if the average global temperature in 2050 were as low as ___°F"). Then participants gave ratings of concern about the projection (on a 6-point scale, from "not at all" to "extremely") and trust in scientists' estimate (on a 6-point scale, from "not at all" to "completely").

2.1.4. *Design*

There were three fully crossed independent variables: stated target year (2015, 2025, 2050, 2100), projection format (deterministic, uncertainty), and place (global and local). All three variables were manipulated between groups so that each participant was randomly assigned to one of 16 conditions.

2.2. *Results*

The analyses reported below as well as those in the following experiment exclude participants who might not have been taking the task seriously. Participants were excluded whose estimates were more than four times the projected values or whose lower bound estimates were greater than their upper bound estimates. We also eliminated all minors.

Applying those exclusion criteria left 780 participants (30.9% female, mean age = 49.29 years).

We predicted greater concern about near- than far-future climate change, greater concern about local than global climate change, and greater trust and concern when projections included uncertainty estimates. Although our hypothesis about the effect of uncertainty estimates was confirmed, we found *no effect* for year and the *opposite* effect for place (greater concern and trust for global projections²). We conducted two ANOVAS on concern (for temperature and sea level) and two on trust (for temperature and sea level). There were three independent variables in each: stated target year (2015, 2025, 2050, and 2100), place (global, local), and format (uncertainty, deterministic). For temperature concern ratings, there was greater concern for global than local projections, $F(1, 779) = 24.98, p < .01$, Cohen's $d = 0.35$, and greater concern for the uncertainty than for deterministic projections, $F(1, 779) = 3.95, p = .05$, Cohen's $d = .14$. There were also two significant interactions. The time by place interaction, $F(3, 779) = 3.74, p = .01$, indicated that the greatest difference in concern between global and local projections was for near-future projections (2015) and the difference declined thereafter. The time by format interaction, $F(3, 779) = 3.10, p = .03$, indicated that format made the greatest difference for projections in the near (2015) and distant future (2100). The analysis of sea level concern ratings was similar. There was greater concern for global than local projections, $F(1, 779) = 16.50, p < .01$, Cohen's $d = 0.29$, but none of the other main effects or interactions reached significance. See Tables 1, 2, and 3.

For temperature trust ratings, there was significantly greater trust in global than local projections, $F(1, 779) = 8.27, p < .01$, Cohen's $d = 0.20$, and significantly greater trust

Table 1
Mean ratings of trust (from 1, “not at all,” to 6, “completely”) and concern (from 1, “not at all,” to 6, “extremely”) for temperature and sea level projections by target year (standard deviation in parentheses)

Target Year	Temperature Concern	Sea Level Concern	Temperature Trust	Sea Level Trust
2015	3.44 (1.70)	3.54 (1.64)	3.59 (1.42)	3.51 (1.45)
2025	3.55 (1.43)	3.58 (1.46)	3.76 (1.21)	3.58 (1.32)
2050	3.85 (1.56)	3.80 (1.62)	3.84 (1.28)	3.84 (1.29)
2100	3.55 (1.46)	3.58 (1.46)	3.67 (1.32)	3.62 (1.25)
<i>Overall</i>	3.60 (1.54)	3.63 (1.55)	3.72 (1.31)	3.64 (1.33)

Table 2
Mean ratings of trust (from 1, “not at all,” to 6, “completely”) and concern (from 1, “not at all,” to 6, “extremely”) for temperature and sea level projections by place (standard deviation in parentheses)

Place	Temperature Concern	Sea Level Concern	Temperature Trust	Sea Level Trust
Global	3.87 (1.55)	3.85 (1.52)	3.85 (1.31)	3.75 (1.33)
Local	3.34 (1.49)	3.41 (1.54)	3.59 (1.29)	3.53 (1.33)
<i>Overall</i>	3.60 (1.54)	3.63 (1.55)	3.72 (1.31)	3.64 (1.33)

Table 3

Mean ratings of trust (from 1, “not at all,” to 6, “completely”) and concern (from 1, “not at all,” to 6, “extremely”) for temperature and sea level projections by projection format (standard deviation in parentheses)

Projection Format	Temperature Concern	Sea Level Concern	Temperature Trust	Sea Level Trust
Deterministic	3.49 (1.58)	3.56 (1.57)	3.61 (1.29)	3.51 (1.33)
Uncertainty	3.71 (1.50)	3.70 (1.52)	3.82 (1.32)	3.77 (1.32)
<i>Overall</i>	3.60 (1.54)	3.63 (1.55)	3.72 (1.31)	3.64 (1.33)

in the uncertainty than in the deterministic format, $F(1, 779) = 4.93$, $p = .03$, Cohen's $d = 0.16$. Finally, there was an interaction between time and place, $F(3, 779) = 3.36$, $p = .02$, indicating that the greatest difference in trust between global and local projections was for near-future projections and the difference declined thereafter (following the concern pattern). The analysis of sea level trust ratings was similar, with greater trust in global than local projections, $F(1, 779) = 6.01$, $p = .01$, Cohen's $d = 0.17$, and greater trust in the uncertainty than deterministic format, $F(1, 779) = 7.19$, $p < .01$, Cohen's $d = 0.20$. There was also an interaction between time and place, $F(3, 779) = 4.32$, $p < 0.01$, indicating that the greatest difference in trust between global and local projections was for near-future projections (following the concern pattern). See Tables 1, 2, and 3.

Thus, the results of Experiment 1 suggest that uncertainty information increases trust and concern for climate change projections. However, surprisingly there was no effect of target year on either concern or trust. Moreover, there was greater concern for and trust in global than local changes. These results were similar to those of the pilot study in which projection values varied naturally.

Participants' understanding of climate projections might also be reflected in their own numeric estimates for temperature and sea level. On average, participants provided estimates that were lower than the projections they were shown, except for sea level in 2100 (Fig. 1). Moreover, participants' change values were larger for target years further in the future and for global projections, despite the fact that the projection values were identical in all conditions. ANOVAS were conducted on temperature and sea level estimates with three independent variables: stated target year (2015, 2025, 2050, 2100), place (global, local), and format (uncertainty, deterministic). Analysis of temperature estimates revealed a significant main effect for target year, $F(3, 779) = 7.26$, $p < .01$, with larger changes estimated farther in the future. In addition, global change estimates ($M = 2.65^{\circ}\text{F}$ [1.47°C]) were significantly larger than were local change estimates ($M = 2.45^{\circ}\text{F}$ [1.36°C]), $F(1, 779) = 4.39$, $p = .04$, Cohen's $d = 0.15$. A similar pattern was detected in the analysis of sea level estimates, with larger changes estimated further in the future, $F(3, 779) = 14.82$, $p < .01$, and for global ($M = 5.13$ in. [13.03 cm]) as compared to local ($M = 4.63$ in. [11.76 cm]), $F(1, 779) = 7.48$, $p < .01$, Cohen's $d = 0.15$. No other main effects or interactions were significant. Thus, there was an effect of both time and place on participants' numeric estimates, even though the projections were identical in all conditions in this experiment (Fig. 1).

Next, we subtracted “as-low-as” from “as-high-as” estimates to create range estimates. On average, temperature ranges were slightly larger than the 2°F (1.11°C) predictive intervals provided in the uncertainty condition. Sea level ranges were slightly smaller than the 6-inch (15.24 cm) predictive intervals (Fig. 2). Moreover, for both temperature and sea level, participants expected an increase in range with time. ANOVAS were conducted on temperature and sea level ranges with three independent variables: target year (2015, 2025, 2050, 2100), place (global, local), and format (uncertainty, deterministic). There was a significant main effect of time for both temperature, $F(3, 779) = 3.96$, $p < .01$, and sea level, $F(3, 779) = 15.30$, $p < .01$. Ranges increased with year (Fig. 2), suggesting that participants anticipated greater uncertainty farther in the future. The only other significant effect was a time by format interaction for sea level, $F(3, 779) = 5.29$, $p < .01$. While those in the deterministic condition anticipated wider ranges with time, the effect was attenuated in the uncertainty condition in which participants were given

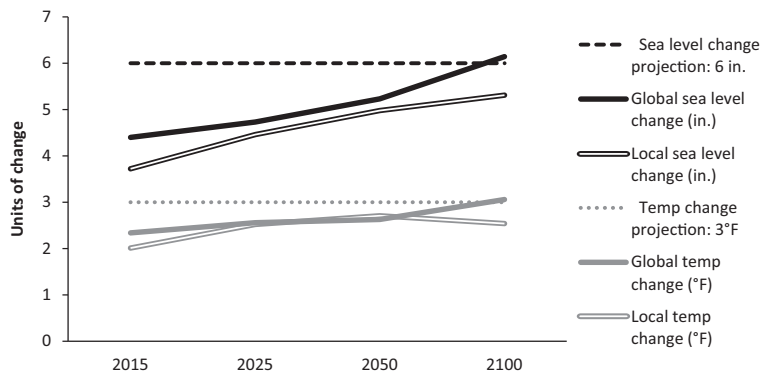


Fig. 1. Participants' numeric estimates of global and local temperature and sea level change by target year. Dotted lines indicate the projections participants were shown, which were held constant across target year.

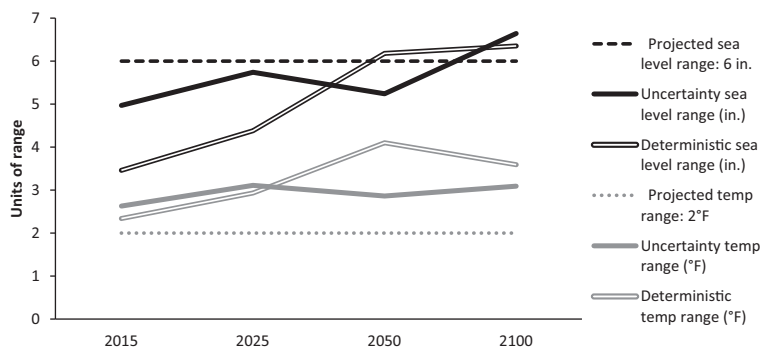


Fig. 2. Deterministic and uncertainty participants' range estimates (difference between upper and lower bound estimates) of temperature and sea level change by stated target year. Dotted lines indicate the projection ranges participants were shown, which were held constant across target year.

6-inch 90% predictive intervals for all target years. Their ranges remained closer to six inches overall, larger than in the deterministic condition in the first two years and smaller in the last two years.

2.3. Discussion

There was no evidence for temporal discounting in Experiment 1, suggesting that concern about climate change is not susceptible to temporal discounting effects. This confirmed results of a pilot study conducted on university undergraduates, using the same basic procedures but providing projection values that increased naturally with year. In the pilot study as well, there was no effect of year on concern.³

It is interesting to note, however, that in Experiment 1 there was an impact of time on participants' numeric estimates. Participants' numeric estimates *increased* with target year, despite the fact that projected values were identical, suggesting that participants understood that the climate would change over time. A similar increase with target year was observed in range estimates, suggesting the participants also anticipated greater uncertainty in the more distant future.

Contrary to the personal relevance hypothesis, there was greater concern for and trust in global as compared to local projections despite the fact the projection values were held constant. A similar pattern was found in the pilot study. Perhaps participants were engaging in comparative optimism, in which one tends to regard local conditions as more favorable (Gifford et al., 2009). They may also have believed that major effects of climate change will be felt elsewhere in the immediate future and locally only later (Lorenzoni & Pidgeon, 2006). Indeed, participants estimated greater global change values for both temperature and sea level. Moreover, several significant interactions suggested that this effect was greatest in the near future, for which global projections inspired much greater trust and concern than did local projections. Perhaps this particular sample, recruited from a weather blog, understood that the effects of climate change in the Pacific Northwest of the United States would be minimal in the near future in contrast to other locations. It may have been this understanding that inspired greater concern for global as opposed to local projections.

Most important, in Experiment 1 we found that, as with weather forecasts, people have greater trust in climate projections that include an uncertainty estimate. This may have been due to the fact that, as with weather forecasts, people had intuitions about the uncertainty involved. Thus, acknowledging the uncertainty in the projection itself may have made it seem more trustworthy.

However, it is important to note that the participant sample used in Experiment 1 was not representative of the U.S. population: It was disproportionately male (69.1%), older (mean age = 49.29 years), and most respondents were from the Puget Sound area in Washington State (93.4%). Moreover, respondents were readers of a blog about weather and climate and may have had a richer understanding of the issues. To assess the generalizability of these results, a more representative sample was required.

3. Experiment 2: M-Turk participants

Experiment 2 was conducted to determine whether the effects of uncertainty estimates extended to a more diverse sample of participants with a wide range of ages, education levels, and political affiliations. This was important because uncertainty information may have a different effect on some groups. For instance, people with less education may misinterpret or mistrust uncertainty information, perhaps due to low numeracy (e.g., Cavanaugh et al., 2008). Moreover, people may interpret information differently due to prior beliefs, especially those associated with political affiliation (Druckman, Peterson, & Slothuus, 2013; Hart & Nisbet, 2012; McCright & Dunlap, 2011). We may find that uncertainty estimates increase trust among some people but decrease trust among others, a “boomerang effect” (Hart & Nisbet, 2012), which may in turn lead to reduced concern. Therefore, Experiment 2 participants were recruited through Amazon’s M-Turk, a portal to surveys that participants complete for small monetary rewards. Evidence suggests that M-Turk samples are reliable and reasonably representative of the general public (Paolacci, Chandler, & Ipeirotis, 2010). In addition, Experiment 2 focused solely on the effects projection format. Time and place were held constant.

3.1. Method

3.1.1. Procedure

An online questionnaire was linked to Amazon’s Mechanical Turk (M-Turk) from August to October of 2013. Participants provided information about their age, gender, and zip code. They selected the category⁴ that best described their educational attainment (high school diploma or less, technical school/some college, college degree, post-graduate work or degree, other), and political party affiliation (Republican, Democrat, independent, unaffiliated, other). Then they completed the questionnaire, which took about 3 minutes.

3.1.2. Participants

A total of 1,015 people (56.7% female, mean age = 33.42) participated and were paid about \$.05 each. Compared to the national population, the M-Turk sample was slightly younger and more female. The sample had a slightly higher proportion of Democrats and was better educated (see Table 4). Nonetheless the M-Turk sample was considerably more similar to the national population than was the weather blog sample used in Experiment 1.

3.1.3. Questionnaire

Each questionnaire showed climate projections for 2050 identical to those used in Experiment 1: a global average temperature increase of 3°F (1.67°C) and a global average sea level rise of 6 inches (15.24 cm). However, this time they were only identified as global averages for 2050 as time and place were held constant. Half of the questionnaires

also included 90% predictive intervals (temperature increase: 2–4°F [1.11–2.22°C], sea level rise: 3–9 inches [7.62–22.86 cm]). After each projection, respondents provided their own single-value numeric estimate. They selected one of three options and filled in the corresponding blank (Fig. 3).

Participants provided lower and upper bound estimates as before. Finally, they indicated their trust in scientists’ estimate (on a 6-point scale, from “not at all” to “completely”) and concern about the projection (on a 6-point scale, from “not at all” to “extremely”).

3.1.4. Design

The experiment had one between-participants independent variable, projection format, with two levels: deterministic and uncertainty.

3.2. Results

The analyses reported below used the same exclusion criteria as in the first experiment. We also eliminated those who selected more than one option concerning future parameter value changes (e.g., temperature will increase by X°F *and* temperature will decrease by Y°F) and all minors, leaving 787 participants.

Table 4
Comparison of M-Turk sample and national population demographics

	M-Turk Sample	M-Turk Sample After Exclusion Criteria Applied	National Population (2012 census data)
Age	Median 30 years	Median 30 years	Median 37.4 years
Gender	56.7% female	55.9% female	50.8% female
Political party ^a	Republican 17.2%, Democrat 32.4%, Independent/other/ unaffiliated 50.3%	Republican 17.4%, Democrat 32.7%, Independent/other/ unaffiliated 49.9%	Republican 24%, Democrat 29%, Independent 45% (2% unidentified) ^a
Education	56.7% college educated +	60.5% college educated +	29.1% college educated +
Zip code	All states represented	All states represented	N/A

Note. ^a<http://www.gallup.com/poll/15370/party-affiliation.aspx>

I think that ...

a. the average global temperature by 2050 will be ____ °F *greater* than the 20th century average.

b. the average global temperature by 2050 will be ____ °F *less* than the 20th century average.

c. the average global temperature by 2050 will be the *same* as the 20th century average.

Fig. 3. Options for the questionnaire projection estimates.

We predicted that the inclusion of uncertainty estimates would increase participants' trust in climate change projections and perhaps concern as well. Moreover, we were particularly interested in whether prior beliefs or education interacted with the predicted effects of uncertainty estimates. To test these questions, we conducted four ANOVAS. The dependent variables were trust and concern ratings for temperature and sea level projections.

For the first two ANOVAS, the independent variables were projection format (uncertainty, deterministic) and a new variable created by classifying participants as either "believers," those who indicated an increase in global average projection values, or "doubters," those who did not. There was greater trust in the uncertainty condition than in the deterministic condition for both temperature, $F(1, 698) = 7.01, p < .01$, Cohen's $d = .12$, and sea level, $F(1, 698) = 5.89, p = .02$, Cohen's $d = .12$. There was also an effect of believer/doubter category on trust for both temperature, $F(1, 698) = 307.87, p < .01$, Cohen's $d = 1.82$, and for sea level, $F(1, 698) = 299.72, p < .01$, Cohen's $d = 1.80$, with greater trust among believers than doubters. There was also an effect of believer/doubter category on concern for both temperature, $F(1, 698) = 199.75, p < .01$, Cohen's $d = 1.61$, and sea level, $F(1, 698) = 253.77, p < .01$, Cohen's $d = 1.78$, with greater concern among believers than doubters (Fig. 4). Although none of the interactions reached significance, surprisingly they trended in the opposite direction to that predicted (greater increase in trust and concern with uncertainty format for doubters).

The next two ANOVAS, with format (uncertainty, deterministic) and a dichotomized education variable (college education or more, high school education or less) as the independent variables, showed significantly greater concern among the college-or-more group as compared to the high-school-or-less group for temperature, $F(1, 786) = 6.64, p = .01$, Cohen's $d = 0.19$, and for sea level, $F(1, 786) = 5.48, p = .02$, Cohen's $d = 0.17$. In addition, there was a significant interaction between education and format for temperature trust, $F(1, 786) = 4.05, p = .05$. While trust ratings between the two levels of education

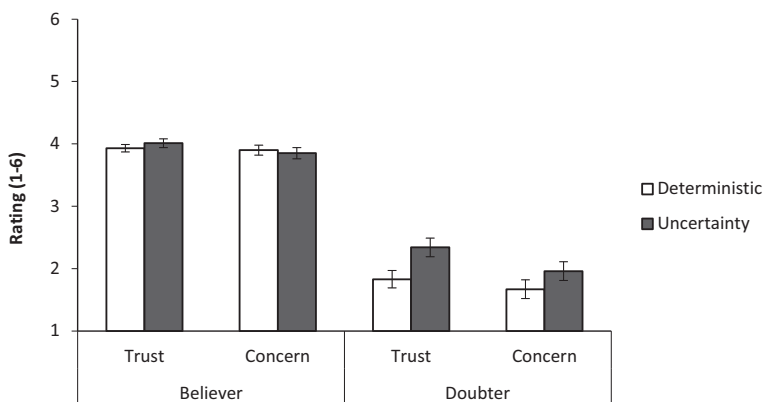


Fig. 4. Combined temperature and sea level trust ratings (from 1, "not at all," to 6, "completely") and concern ratings (from 1, "not at all," to 6, "extremely") by believer/doubter and projection format.

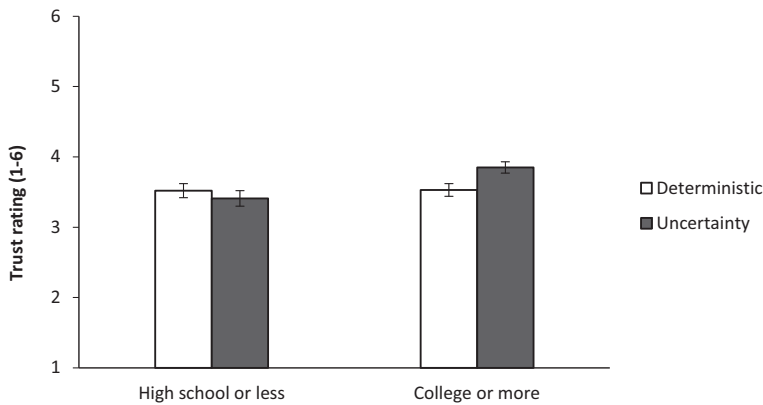


Fig. 5. Temperature trust ratings (from 1, “not at all,” to 6, “completely”) by education level and projection format.

were similar in the deterministic condition, those with a college education or higher had greater trust in the projection that included the uncertainty estimate than did those with a high school education or less. Moreover, those with a high school education or less trusted the two formats approximately equally (Fig. 5).

These analyses suggest that the main impact of uncertainty estimates is to increase rather than decrease trust, although this impact may depend on level of education. Importantly, the inclusion of uncertainty estimates did not result in a significant decrease in trust or concern for either group. The next analyses demonstrate that predictive intervals influenced the *amount* of uncertainty participants anticipated as well. We subtracted the “as-low-as” from the “as-high-as” estimates among those who anticipated an increase in projection values in 2050⁵ to create range estimates. Although temperature ranges were generally larger than the 2°F (1.11°C) provided in the predictive interval, they were significantly smaller in the uncertainty condition ($M = 3.43^{\circ}\text{F}$ [1.90°C], $SD = 2.24^{\circ}\text{F}$ [1.24°C]) than in the deterministic condition ($M = 4.05^{\circ}\text{F}$ [2.25°C], $SD = 2.61$ [1.45°C]), suggesting that people were influenced by the uncertainty estimate, $t(629) = 3.21$, $p < .01$, Cohen’s $d = 0.25$. Sea level ranges were generally smaller than the 6-inch range provided in the uncertainty condition, and although the ranges estimated by those in the uncertainty condition were larger, the difference did not reach statistical significance.

We also predicted that in this more diverse sample, differences in trust and concern due to demographic variables might be detected. Indeed, age was significantly negatively correlated with both trust ($r = -.21$, $p < .01$) and concern ($r = -.11$, $p < .01$) ratings (Fig. 6). The older respondents were, the less concern and trust they reported. Age was also related to belief in climate change. We conducted a chi-square test of independence to examine the relationship between believer and doubter category and a dichotomized age variable created by conducting a median split on age. Thus, the null hypothesis for this analysis was that participants’ belief in climate change was independent of their age. To the contrary, however, there was a greater proportion of participants over 30 who

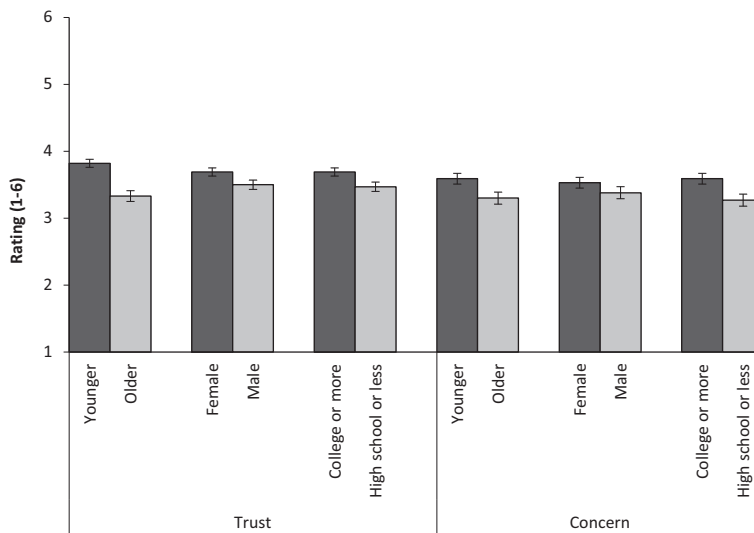


Fig. 6. Combined temperature and sea level trust ratings (from 1, “not at all,” to 6, “completely”) and concern ratings (from 1, “not at all,” to 6, “extremely”) by age, gender, and education level.

thought there would be no increase in global average temperature (26.09%, $\chi^2(1) = 20.83$, $p < .01$), or sea level (25.00%, $\chi^2(1) = 11.03$, $p < .01$) than of participants 30 years and under (global average temperature, 13.16%; or sea level, 15.48%), suggesting that older individuals were less likely to believe in climate change than were younger individuals.

Gender made a difference as well. Men reported significantly less trust in temperature projections, $t(713.07) = 2.30$, $p = .02$, and significantly less concern for sea level projections, $t(785) = 2.18$, $p = .03$, than did women (Fig. 6). However, a chi-square analysis testing whether belief in climate change was independent of gender revealed that approximately equal proportions of men and women thought there would be no increase in global average temperature (women, 19.1%; men 20.5%; $\chi^2(1) = 0.16$, $p = .69$) and sea level (women, 20.0%; men 20.7%; $\chi^2(1) = 0.10$, $p = .75$). Thus, the gender difference was restricted to trust and concern, rather than extending to the belief that the global climate was changing.

Educational differences had an impact on concern. Those with a high school education or less expressed less concern for temperature change, $t(785) = 2.58$, $p = .01$, Cohen's $d = 0.19$, and sea level rise, $t(785) = 2.34$, $p = .02$, Cohen's $d = 0.17$, than did those with a college or advanced degree (Fig. 6). However, a chi-square testing whether belief in climate change was independent of education revealed that approximately equal proportions of high school-or-less and college-or-more participants thought there would be no increase in global average temperature (high school or less, 20.7%; college or more, 18.8%; $\chi^2(1) = 0.40$, $p = 0.53$) and sea level (high school or less, 21.1%; college or more 19.7%; $\chi^2(1) = 0.24$, $p = .62$).

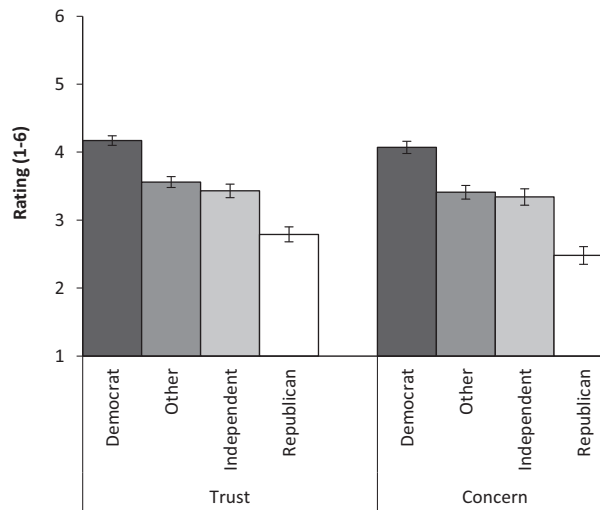


Fig. 7. Combined temperature and sea level trust ratings (from 1, “not at all,” to 6, “completely”) and concern ratings (from 1, “not at all,” to 6, “extremely”) by political party affiliation.

The largest overall difference was due to political party affiliation. ANOVAS with political party (Republican, Democrat, Independent, other) as the independent variable revealed effects on temperature trust, $F(3, 786) = 45.95$, $p < .01$, sea level trust, $F(3, 786) = 45.04$, $p < .01$, temperature concern, $F(3, 786) = 41.33$, $p < .01$, and sea level concern, $F(3, 786) = 33.99$, $p < .01$. As can be seen in Fig. 7, Democrats expressed the greatest trust and concern, Republicans the least, and the other groups fell in between. Political party had an impact on belief in climate change, as well. Fewer Republicans believed that global temperature (62.0%) or global sea level (65.0%) would increase by 2050 than did Democrats (temperature 90.7%, sea level 87.2%) or Independents and others (temperature 79.9%, sea level 79.9%).

3.3. Discussion

Experiment 2, using a more representative sample than in the previous experiment, revealed a striking benefit of including uncertainty estimates: Participants who were shown climate projections including 90% predictive intervals trusted them significantly more overall than did participants who were shown single-value projections. Importantly, we found no evidence in any analysis that participants who were shown predictive intervals trusted the projections less. Moreover, the predictive interval had an impact on the amount of uncertainty the participants anticipated. Those whose projection included a predictive interval expected significantly smaller temperature ranges than those in the deterministic condition.

In addition, this experiment demonstrated that demographic characteristics have an important impact on trust and concern with regard to climate projections, and in some

cases on belief in global climate change. More than a third of older respondents and more than half of those who identify themselves as Republicans indicated that they believed there would be no increase in global average temperature and sea level by 2100, suggesting that they do not believe in climate change at all. Not surprisingly, these groups expressed less trust in and concern about climate projections.

However, there were group differences that were reflected in trust and concern alone, with no differences in belief in global warming. This was the case with gender. Men expressed less trust and concern than did women, but approximately 80% of both groups expressed belief in global warming. Similarly those with a high school degree or less were both less trusting and less concerned, but again approximately 80% of both education groups expressed belief in global warming.

Importantly, the predicted interaction between demographic characteristics and projection format was not observed. We were concerned that projections with uncertainty estimates might provide those who were already skeptical about global warming additional fuel for skepticism, resulting in a boomerang effect, that is, decreased trust when the 90% predictive interval was included. This was clearly not the case. In fact, the trend was in the opposite direction, showing a greater increase in trust and concern among doubters than believers when the predictive interval was included. Thus, the only impact of the uncertainty estimate was to *increase* trust across groups, with the exception of those with a high school degree or less. Even with this group, however, it did not significantly decrease trust. It simply failed to increase trust, perhaps because they had difficulty understanding what it meant.

4. General discussion

This pair of experiments suggests that including uncertainty estimates in climate projections generally increases trust in them. It may be that most people already understand that climate prediction involves uncertainty, regardless of whether they have prior experience of the relationship between the projection and the outcome. Indeed, the majority of participants, even those with deterministic projections, anticipated a range of outcomes. Moreover, participants' ranges increased with target year in Experiment 1 despite the uniform predictive intervals, suggesting that participants also anticipated an increase in uncertainty with lead-time. Thus, because of these preexisting intuitions, users may have greater trust in climate projections that acknowledge and quantify the uncertainty, as with weather forecasts. The predictive intervals also influenced participant understanding of the amount of uncertainty involved, increasing range estimates for sea level in Experiment 1 and reducing them for temperature in Experiment 2. Moreover, enhancing trust in climate change projections may over time with multiple exposures lead to an increase in concern, for which we saw suggestive evidence in Experiment 1 reported here, and eventually greater commitment to climate-friendly behavior.

This research also provides insight into concern about climate change. We found no evidence for temporal discounting (less concern for future events) either in Experiment 1,

with identical change values for all target years, or in a pilot study, with change values that increased with target year. It is important to note that either method introduces a potential confound. Identical change values may imply a slower rate of change for years in the more distant future and may lead to less concern for that reason. Naturally increasing change values may lead to greater concern in later years because they describe more change. However, both methods resulted in concern that remained statistically equivalent across the years. Although this is a null finding, it may well demonstrate an important principle regarding concern for environmental issues like climate change: People may recognize that the well-being of the environment is equally important in the future as it is in the present (Böhm & Pfister, 2005; Gattig & Hendrickx, 2007). This is clearly an area that requires further research and may have important implications for climate communication. Greater concern for global than local change, found both in the pilot study and Experiment 1, may be due to a similar understanding.

Experiment 2 also revealed an interesting dissociation between trust and concern on one hand and belief in climate change on the other. Men and those with less education expressed less trust and concern, but approximately 80% were believers in all groups. However, for age and political affiliation, these variables were aligned. There were larger proportions of doubters among older people and Republicans, and both groups expressed less trust and concern than did younger people and people with other political affiliations. These intriguing results point to a potentially fruitful avenue for future research exploring the relationship between these variables and motivation for climate-friendly behavior. Lack of trust and concern may impact decisions to engage in climate-friendly behavior, to the same or to a lesser degree than lack of belief. Moreover, communication strategies that motivate behavior may be different depending on these underlying factors.

Perhaps the most important finding in these experiments concerns the expression of uncertainty information. Many fear that acknowledging uncertainty in climate projections will backfire, decreasing trust by giving rise to the impression that climate scientists are divided or incompetent. That was clearly not the case here. Although the effect sizes were small, they were remarkably consistent. In every case, projections that included an uncertainty estimate were regarded as significantly more trustworthy overall than those that did not. Moreover, it is important to note that this was the result of a single exposure. If the majority of users' exposure to climate information took this form, the impact may well be much greater. Furthermore, uncertainty estimates increased rather than decreased trust even among subgroups whose beliefs were contradictory to climate change. Only among those with a high school education or less was no increase in trust observed. For this group, uncertainty and deterministic projections were trusted approximately equally. It is possible that this group did not understand the uncertainty format used here. Perhaps another expression would be more accessible. This is clearly an avenue for future work.

However, for the majority of participants in both experiments, the effect of uncertainty information was to increase trust in climate projections. It may well be that everyday users have more sophisticated intuitions about uncertainty than they are generally given credit for. Thus, predictions that include an explicit estimate of uncertainty may seem more plausible to most people. Scientists are often privy to reliable uncertainty estimates

in many domains that have important implications for public safety and well-being. However, scientists are often reluctant to share this information for fear it will be confusing to the general public or reduce credibility (Frewer et al., 2003; Shackley et al., 1999). This research suggests that such fears may be unfounded. In fact, clearly expressed uncertainty estimates may have a positive impact on both users' understanding of potential future events as well as their trust in the scientific source of that information. For issues as weighty as climate change, it may be time to take advantage of this important communication strategy.

Acknowledgments

This material is based on work supported by the National Science Foundation grant 1430781.

Notes

1. Although predictive intervals are not yet reliably calculable for climate projections, we used theoretical interval values here because this is among the main goals of climate scientists.
2. This was true regardless of whether or not the sample was restricted to the 93% who claimed to live in the Puget Sound area.
3. Tested in a series of one-way ANOVAs on concern ratings for global temperature, $F(3, 269) = 0.15$, $p = .93$, global sea level, $F(3, 267) = 0.64$, $p = .59$, local temperature, $F(3, 269) = 0.06$, $p = .98$, and local sea level, $F(3, 268) = .64$, $p = .59$., with target year (2015, 2025, 2050, and 2100) as the independent variable.
4. Categories used by Gallup.
5. Only 17 participants estimated a temperature decrease and it was on average close to the 3°F (1.66°C) *increase* value in the projections ($M = 3.21^\circ\text{F}$ [1.78°C], $SD = 2.20^\circ\text{F}$ [1.22°C]), so we are hesitant to draw any conclusions as these participants may have misunderstood the question.

References

- Ainslie, G. (1975). Specious reward: A behavioral theory of impulsiveness and impulse control. *Psychological Bulletin*, 82(4), 463–509.
- Baron, J. (2000). Can we use human judgments to determine the discount rate? *Risk Analysis*, 20(6), 861–868.
- Böhm, G., & Pfister, H. R. (2005). Consequences, morality, and time in environmental risk evaluation. *Journal of Risk Research*, 8(6), 461–479.
- Budescu, D. V., Por, H. H., & Broomell, S. B. (2012). Effective communication of uncertainty in the IPCC reports. *Climatic Change*, 113(2), 181–200.

- Cavanaugh, K., Huizinga, M. M., Wallston, K. A., Gebretsadik, T., Shintani, A., Davis, D., & Rothman, R. L. (2008). Association of numeracy and diabetes control. *Annals of Internal Medicine*, 148(10), 737–746.
- Center for Research on Environmental Decisions (CRED) (2009). *The psychology of climate change communication: A guide for scientists, journalists, educators, political aides, and the interested public*. New York.
- Chapman, G. B. (2002). Your money or your health: Time preferences and trading money for health. *Medical Decision Making*, 22(5), 410–416.
- Cook, J., Nuccitelli, D., Green, S. A., Richardson, M., Winkler, B., Painting, R., Way, R., Jacobs, P., & Skuce, A. (2013). Quantifying the consensus on anthropogenic global warming in the scientific literature. *Environmental Research Letters*, 8(2), 024024.
- Ding, D., Maibach, E. W., Zhao, X., Roser-Renouf, C., & Leiserowitz, A. (2011). Support for climate policy and societal action are linked to perceptions of scientific agreement. *Nature Climate Change*, 1, 462–466.
- Druckman, J. N., Peterson, E., & Slothuus, R. (2013). How elite partisan polarization affects public opinion formation. *American Political Science Review*, 107(1), 57–79.
- Freudenburg, W. R., & Muselli, V. (2010). Global warming estimates, media expectations, and the asymmetry of scientific challenge. *Global Environmental Change*, 20, 483–491.
- Frewer, L. J., Hunt, S., Brennan, M., Kuznesof, S., Ness, M., & Ritson, C. (2003). The views of scientific experts on how the public conceptualize uncertainty. *Journal of Risk Research*, 6(1), 75–85.
- Gattig, A., & Hendrickx, L. (2007). Judgmental discounting and environmental risk perception: Dimensional similarities, domain differences, and implications for sustainability. *Journal of Social Issues*, 63(1), 21–39.
- Gifford, R., Scannell, L., Kormos, C., Smolova, L., Biel, A., Boncu Corral, V., Hanyu, K., Hine, D., Kaiser, F. G., Korpela, K., Mertig, A. G., Mira, R. G., Moser, G., Passafaro, P., Pedroso, L., Pinheiro, J. Q., Saini, S., Sako, T., Sautkina, E., Savina, Y., Schmuck, P., Schultz, W., Sobeck, K., Sundblad, E., & Uzzell, D. (2009). Temporal pessimism and spatial optimism in environmental assessments: An 18-nation study. *Journal of Environmental Psychology*, 29 (1), 1–12.
- Hardisty, D. J., & Weber, E. U. (2009). Discounting future green: Money versus the environment. *Journal of Experimental Psychology: General*, 138(3), 329–340.
- Hart, P. S., & Nisbet, E. C. (2012). Boomerang effects in science communication: How motivated reasoning and identity cues amplify opinion polarization about climate mitigation policies. *Communication Research*, 39(6), 701–723.
- Joslyn, S., & LeClerc, J. (2012). Uncertainty forecasts improve weather-related decisions and attenuate the effects of forecast error. *Journal of Experimental Psychology: Applied*, 18, 126–140.
- Joslyn, S., & Savelli, S. (2010). Communicating forecast uncertainty: Public perception of weather forecast uncertainty. *Meteorological Applications*, 17, 180–195.
- Leiserowitz, A. (2006). Climate change risk perception and policy preferences: The role of affect, imagery, and values. *Climatic Change*, 77(1–2), 45–72.
- Leiserowitz, A., Maibach, E., Roser-Renouf, C., Feinberg, G., & Howe, P. (2013). *Climate change in the American mind: Americans' global warming beliefs and attitudes in April 2013*. Yale University and George Mason University. New Haven, CT: Yale Project on Climate Change Communication.
- Lewandowsky, S., Gignac, G. E., & Oberauer, K. (2013). The role of conspiracist ideation and worldviews in predicting rejection of science. *PLoS ONE*, 8, e75637.
- Lorenzoni, I., & Pidgeon, N. F. (2006). Public views on climate change: European and USA perspectives. *Climatic Change*, 77(1–2), 73–95.
- McCright, A. M., & Dunlap, R. E. (2011). The politicization of climate change and polarization in the American public's views of global warming, 2001–2010. *The Sociological Quarterly*, 52(2), 155–194.
- Morss, R. E., Demuth, J. L., & Lazo, J. K. (2008). Communicating uncertainty in weather forecasts: A survey of the U.S. public. *Weather and Forecasting*, 23, 974–991.
- Nadav-Greenberg, L., & Joslyn, S. L. (2009). Uncertainty forecasts improve decision making among non-experts. *Journal of Cognitive Engineering and Decision Making*, 3(3), 209–227.

- Paolacci, G., Chandler, J., & Ipeirotis, P. G. (2010). Running experiments on Amazon Mechanical Turk. *Judgment and Decision making*, 5(5), 411–419.
- Roulston, M. S., Bolton, G. E., Kleit, A. N., & Sears-Collins, A. L. (2006). A laboratory study of the benefits of including uncertainty information in weather forecasts. *Weather and Forecasting*, 21, 116–122.
- Savelli, S., & Joslyn, S. (2012). Boater safety: Communicating weather forecast information to high-stakes end users. *Weather, Climate, and Society*, 4(1), 7–19.
- Shackley, S., Risbey, J., Stone, P., & Wynne, B. (1999). Adjusting to policy expectations in climate change modeling. *Climatic Change*, 43(2), 413–454.
- Smithson, M. (1999). Conflict aversion: Preference for ambiguity vs. conflict in sources and evidence. *Organizational Behavior and Human Decision Processes*, 79, 179–198.
- Weber, E. U. (2006). Experience-based and description-based perceptions of long-term risk: Why global warming does not scare us (yet). *Climatic Change*, 77(1–2), 103–120.